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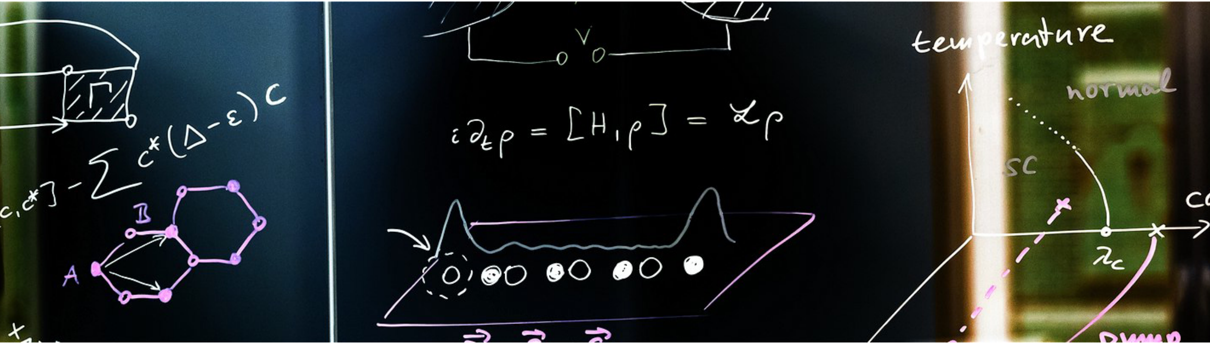
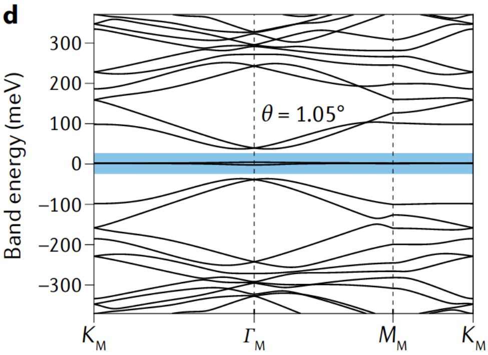
FAKULTÄT
FÜR MATHEMATIK, INFORMATIK
UND NATURWISSENSCHAFTEN

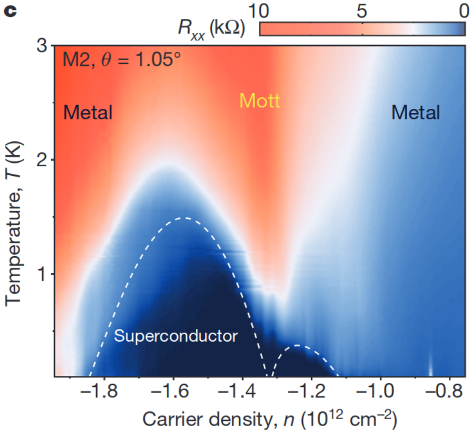


Tjark Sievers – Master's Colloquium May 2025

Superconducting Phase Stiffness and Coherence Length From a Finite Momentum Pairing Constraint

Flat Band Graphene Systems[1,2]





[1] Band structure Törmä, Peotta, and Bernevig, 10.1038/s42254-022-00466-y (*Nature Reviews Physics* 2022)
[2] Cao et al., 10.1038/nature26160 (*Nature* 2018)

Superconductivity in Flat-Band Systems

- Superconductivity: pairing of electrons and phase coherence
- Also enables superconducting current:

$$\mathbf{j} = D_{\mathrm{S}} \mathbf{A} \tag{1}$$

In single band:

$$D_{S,ij} \sim \sum_{\mathbf{k}} \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_i \, \partial k_j} \qquad \text{Vanishes for a flat band!} \tag{2}$$

In multi-band systems[3]:

- Conventional:

$$D_{\mathrm{S},1,ij} \sim \sum_{\mathbf{k}} \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_i \, \partial k_j} \tag{3}$$

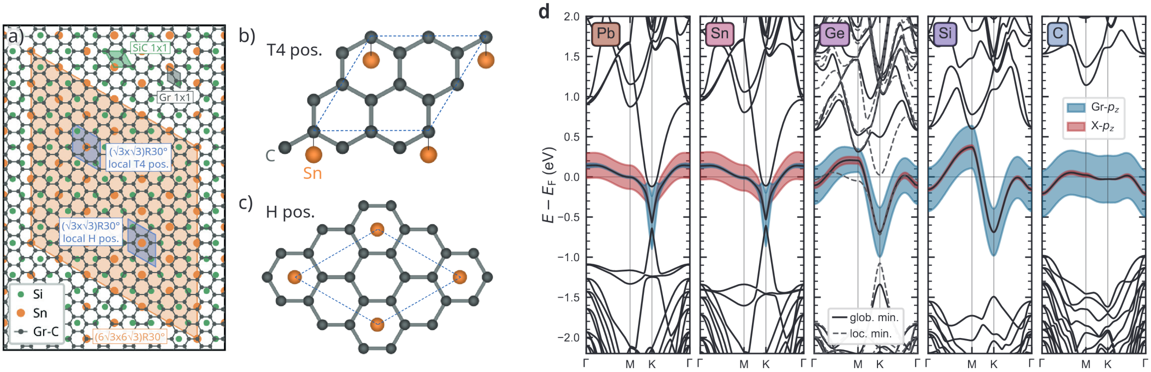
- Geometric:

$$D_{\mathrm{S},2,ij} + D_{\mathrm{S},3,ij} \sim U \mathcal{M}_{ij} \tag{4}$$

  with quantum metric $\mathcal{M}_{ij}$ (the real part of the geometric tensor on the space of Bloch states)

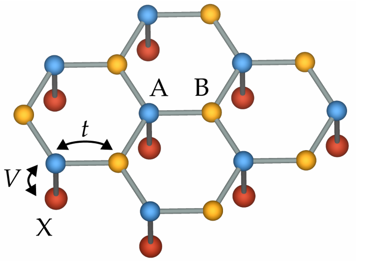
[3] Peotta and Törmä, 10.1038/ncomms9944 (*Nature Communications* 2015)

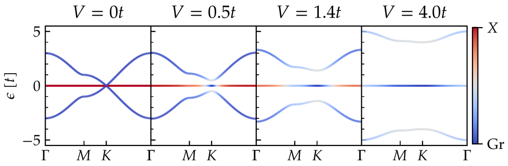
Flat Band by Proximity Coupling[4,5]



[4]Ghosal et al., 10.48550/arXiv.2412.01329 (2024)
[5]Witt et al., 10.48550/arXiv.2503.03700 (2025)

Decorated Graphene Model[6]





- Captures essential elements: graphene lattice, flat band hybridized with graphene dirac cones
- Hamiltonian:

$$H = -t \sum_{\langle ij\rangle,\sigma} c^{(A),\dagger}_{i\sigma} c^{(B)}_{j\sigma} + V \sum_{i\sigma} d^{\dagger}_{i\sigma} c^{(A)}_{i\sigma} + \text{h.c.}$$
$$- \sum_{i,\alpha} U_\alpha c^{\dagger}_{i\alpha\uparrow} c^{\dagger}_{i\alpha\downarrow} c_{i\alpha\downarrow} c_{i\alpha\uparrow} \tag{5}$$

[6] Witt et al. 10.48550/arXiv.2503.03700 (2025)

Ginzburg-Landau Theory

Two ingredients:

- Order parameter:

$$|\Psi| = \begin{cases} 0 & T \geq T_{\mathrm{C}} \\ |\Psi(T)| > 0 & T < T_{\mathrm{C}} \end{cases} . \tag{6}$$

- Free energy: thermodynamic potential, minimized in thermodynamic equilibrium
- Ginzburg-Landau free energy (expansion of free energy in order parameter):

$$f_{\mathrm{GL}}[\Psi, \nabla\Psi] = \frac{\hbar^2}{2m^*}|\nabla\Psi|^2 + r|\Psi|^2 + \frac{u}{2}|\Psi|^4 \tag{7}$$

- Writing $\Psi = |\Psi| e^{\mathrm{i}\phi}$:

$$f_{\mathrm{GL}} = \frac{\hbar^2}{2m^*} |\Psi|^2 (\nabla\phi)^2 + \left[\frac{\hbar^2}{2m^*} (\nabla|\Psi|)^2 + r|\Psi|^2 + \frac{u}{2}|\Psi|^4 \right] \tag{8}$$

- Correlation length:

$$\xi = \sqrt{\frac{\hbar^2}{2m^*|r|}} = \xi_0 \left(1 - \frac{T}{T_{\mathrm{C}}} \right)^{-\frac{1}{2}} \tag{9}$$

  with coherence length $\xi_0 = \xi(T=0) = \sqrt{\frac{\hbar^2}{2m^*|r|T_{\mathrm{C}}}}$
- Coupling to electromagnetic field via vector potential $\mathbf{A}$, i.e. $f_{\mathrm{GL}}[\Psi, \mathbf{A}]$
- Characteristic length scales for variations of $\mathbf{A}$: London penetration depth λ_{L}

Finite-Momentum Pairing Method

- Stationary point of Ginzburg-Landau free energy:

$$|\Psi| = |\Psi_0|\sqrt{1 - \xi^2|\nabla\phi(\mathbf{r})|^2} \tag{10}$$

- Breakdown of order parameter with phase fluctuations $\rightarrow$ correlation length ξ
- Simple choice: $\phi(\mathbf{r}) = \mathbf{q}\cdot\mathbf{r} \implies \xi \propto \frac{1}{q}$
- Finite momentum $\rightarrow$ current $\mathbf{j_q}$, can get

$$\lambda_{\mathrm{L}}(T) \propto \sqrt{\frac{1}{\xi(T)j_{\mathrm{dp}}(T)}}\,, \; D_{\mathrm{S}} \propto \frac{1}{\lambda_{\mathrm{L},0}^2} \tag{11}$$

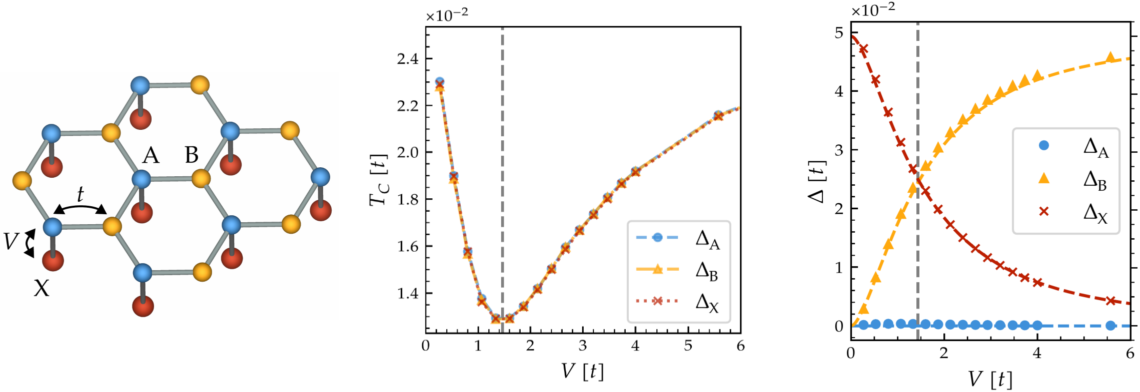
BCS mean field theory

- $H_{\text{int}} \approx \sum_{i,\alpha}\left(\Delta_{i\alpha} c^\dagger_{i\alpha\uparrow} c^\dagger_{i\alpha\downarrow} + \Delta^*_{i\alpha} c_{i\alpha\downarrow} c_{i\alpha\uparrow}\right)$
- Introduce FMP as $\Delta_{i\alpha} = \Delta_\alpha e^{\mathbf{i}\mathbf{q}\mathbf{r}_{i\alpha}}$
- Mean-field Hamiltonian:

$$H_{\text{MF}}(\mathbf{q}) = \sum_{\mathbf{k}} \Psi^\dagger_{\mathbf{q},\mathbf{k}} \begin{pmatrix} H^\uparrow_{\mathbf{k}} - \mu & \Delta(\mathbf{q}) \\ \Delta^\dagger(\mathbf{q}) & -\left(H^\downarrow_{\mathbf{q}-\mathbf{k}}\right)^* + \mu \end{pmatrix} \Psi_{\mathbf{q},\mathbf{k}} \tag{12}$$

with $\Psi_{\mathbf{q},\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}1\uparrow} & c_{\mathbf{k}2\uparrow} & \dots & c_{\mathbf{k}n_{\text{orb}}\uparrow} & c^\dagger_{\mathbf{q}-\mathbf{k}1\downarrow} & c^\dagger_{\mathbf{q}-\mathbf{k}2\downarrow} & \dots & c^\dagger_{\mathbf{q}-\mathbf{k}n_{\text{orb}}\downarrow} \end{pmatrix}$

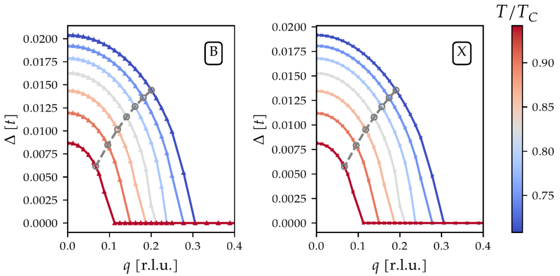
Superconductivity in Decorated Graphene

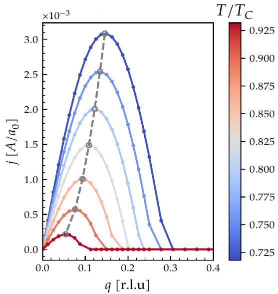


- $U = 0.1t$, results in range of $T_C = 150 - 250\,\mathrm{K}$

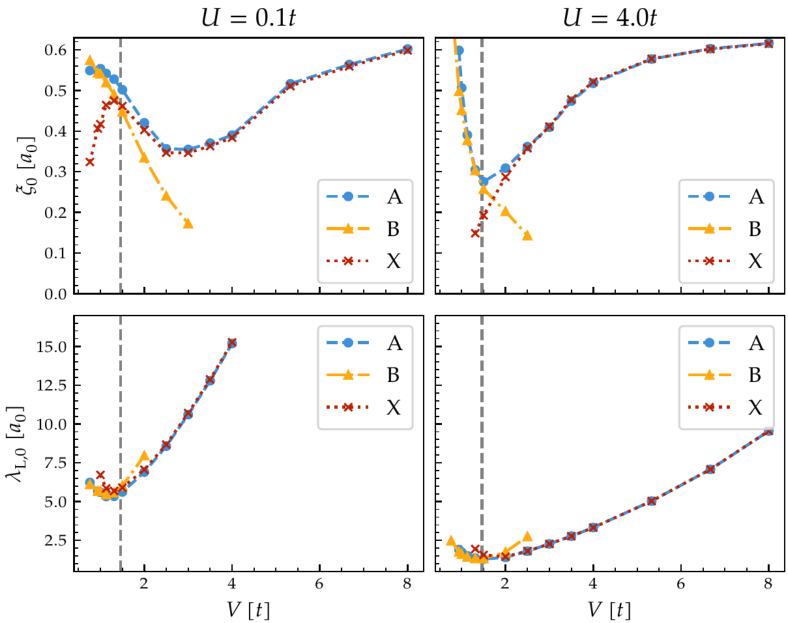
FMP and Length Scales

- Self-consistent calculation under finite $\mathbf{q}$ (both for $V = 1.5t$)

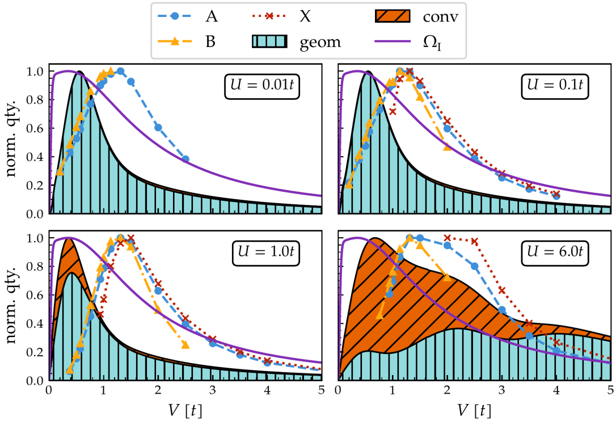




- Marked: points where $^{\Delta}/_{\Delta_0} = {}^{1}/_{\sqrt{2}}$
- Marked: maximum j_{dp}



U = 0.1t
U = 4.0t
ξ0 [a0]
λL,0 [a0]
V [t]
V [t]
A
B
X



- Superfluid weight $D_S \propto \frac{1}{\lambda_{L,0}^2}$
- Quadratic Wannier spread $\Omega_I = \mathrm{Tr}\, \mathcal{M}_{ij}$
- Conventional and geometric contribution calculated from linear response theory[7]

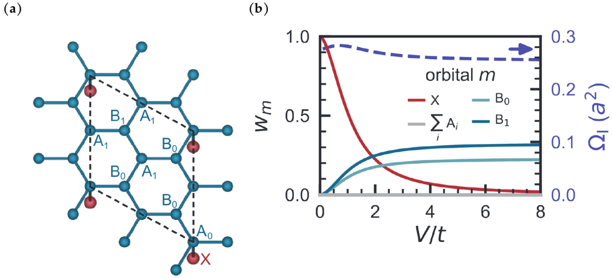
[7] Liang et al., 10.1103/PhysRevB.95.024515 (*Physical Review B* 2017)

Summary

- Decorated graphene: model for graphene adatom heterostructures with flat band hybridized to Dirac cones
- Introduced finite-momentum pairing method to calculate superconducting length scales
- Showed length scales calculated in BCS mean field theory

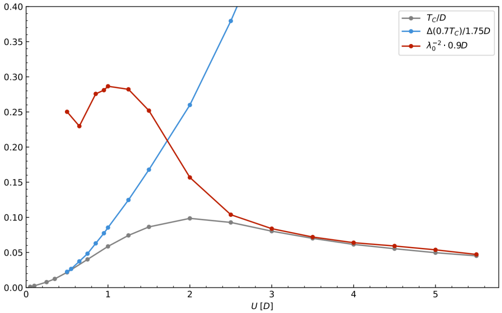
Outlook

- Current project: benchmark FMP method in DMFT for one band Hubbard model
- Compare FMP method in BCS with other results in flat band systems (e.g. Lieb lattice[8] as a standard example), clear up the discrepancies
- Apply FMP method to (more realistic) extended decorated Graphene model



[8] Penttilä, Huhtinen, and Törmä, 10.1038/s42005-025-01964-y (*Communications Physics* 2025)

Results: FMP Method in DMFT (One-Band Hubbard)



- Characterizes what is limiting superconductivity: pairing for low U, phase coherence for high U
- T_C in mean-field theory follows gap